STAFF REVIEW · H446 319 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Work growing with the number of halvings.

2. Condition stopping further calls.

3. Intermediate program states.

4. mid2→5, high1, mid0→1, low1, mid1→3; return index1.

I Do — source worked steps

1. Explain the method — The midpoint is excluded after a failed comparison, so bounds shrink.

We Do — guided CHECK

1. Explain the method
SumTo(2)=2+SumTo(1)=2+1+SumTo(0)=3.

You Do — source mark schemes

1. mid2→6 [2], high1 [1], mid0→2 [1], low1 then mid1→4 [1], return index1 [1].

2. Calls 4,3,2,1,0 [3], returns 0,1,3,6,10 [3].

3. Bubble $O(n^2)$ [1] and O(1) extra [1]; merge O(n log n) [1] and O(n) extra [1]; merge scales better [1] with available memory [1].

Exit Ticket — source answer

1. False. It can repeat the same midpoint.